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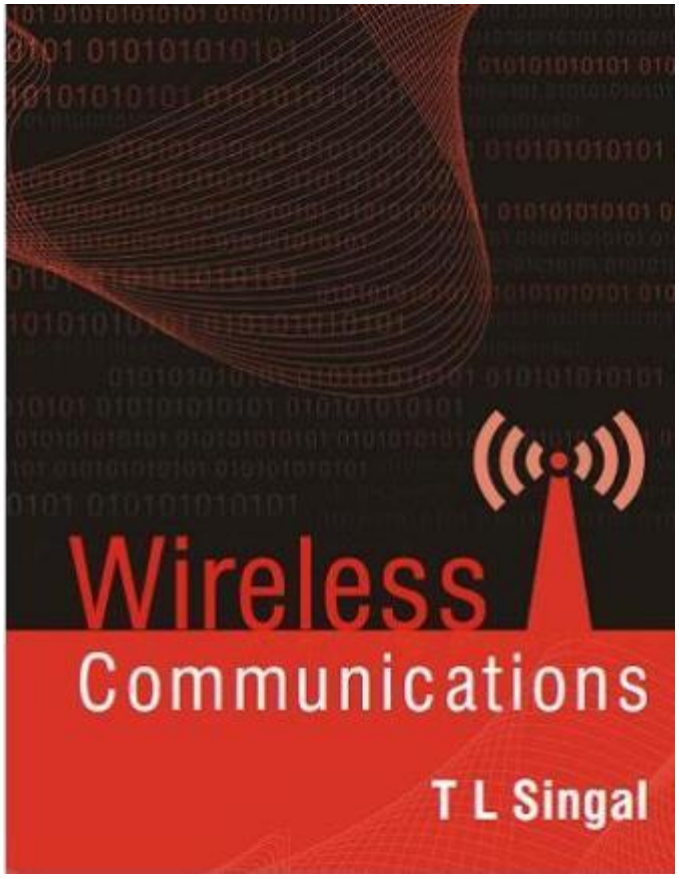

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Evolution of Wireless Communication Systems

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Evolution of Wireless Communication Systems

- Brief History of Wireless Communications
- **Advantages of Wireless Communications**
- Disadvantages of Wireless Communications
- **Wireless Network Generations**
- Comparison of Wireless Systems
- **Evolution to Next Generation Networks**
- Applications of Wireless Communications
- **Potential Market Areas**
- Challenges for Research

What is Wireless Communications?

Wireless Communications – The transmission of user information such as human voice, digital data, e-mail messages, video and other multimedia services without the use of wires

Brief History of Wireless Communications

- Radio and Television Communications
- **Radar Communications**
- **Satellite Communications**
- **Wireless and Mobile Communications**
- **Cellular Communications**

Transition from Analog to Digital Systems

- System capacity
- **Quality aspects**
- Compatibility with other systems such as ISDN

Advantages of Wireless Communications

- ✔ Mobility
- ✔ **Increased reliability**
- ✔ Ease of installation
- ✔ **Rapid disaster recovery**
- ✔ Lower cost

Disadvantages of Wireless Communications

- Radio signal interference
- **Security**
- Health hazards

Wireless Network Generations

- First Generation Analog Cellular Systems
- **Second Generation Digital Cellular Systems**
- Evolution from 2G to 3G Cellular Networks
- **Third Generation Digital Cellular Systems**
- Wireless Networking Technologies

Existing 1G Analog Cellular Systems

- AMPS : Advanced Mobile Phone System
- **ETACS: Enhanced Total Access Communication System**
- NMT : Nordic Mobile Telephone
- **JTACS : Japanese Total Access Communication System**
- NTACS: Narrowband JTACS

Analog Cellular Systems

Standard	Frequency Band	Multiple Access	Modulation	Channel BW
AMPS	**824-894 MHz**	**FDMA**	**FM**	**30KHz**
NAMPS	**824-894 MHz**	**FDMA**	**FM**	**10KHz**
ETACS	900 MHz	FDMA	FM	25KHz
NMT-450	450-470 MHz	FDMA	FM	25KHz
NMT-900	890-960 MHz	FDMA	FM	12.5KHz
JTACS	860-925	FDMA	FM	25KHz

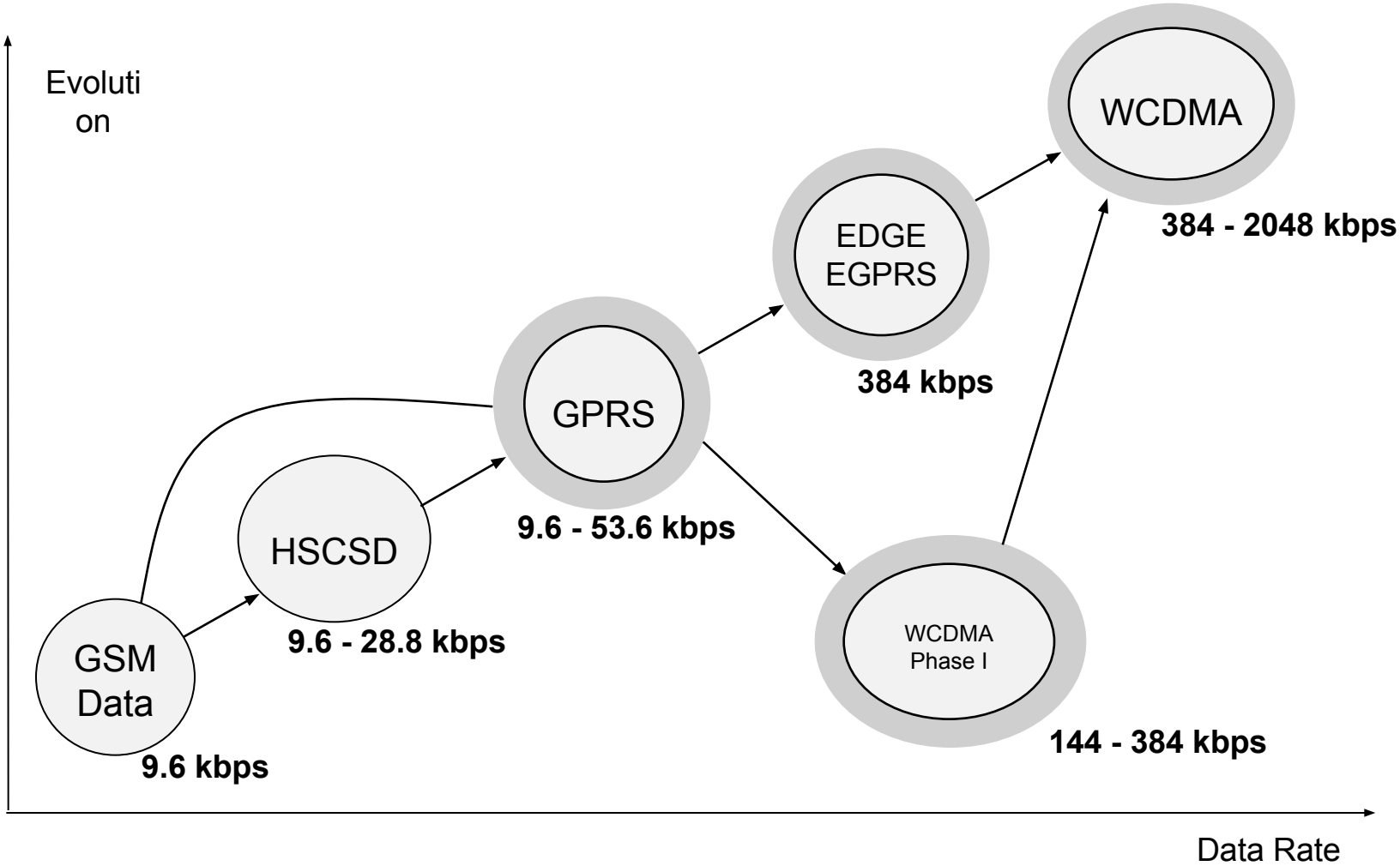
Second Generation Digital Cellular Systems

- IS-54/IS-136 : US Digital Cellular (USDC) - TDMA System
- **GSM : Global System for Mobile**
- PDC : Pacific Digital Cellular - A Japanese TDMA Cellular Standard
- **IS-95/cdmaOne : CDMA Cellular System**

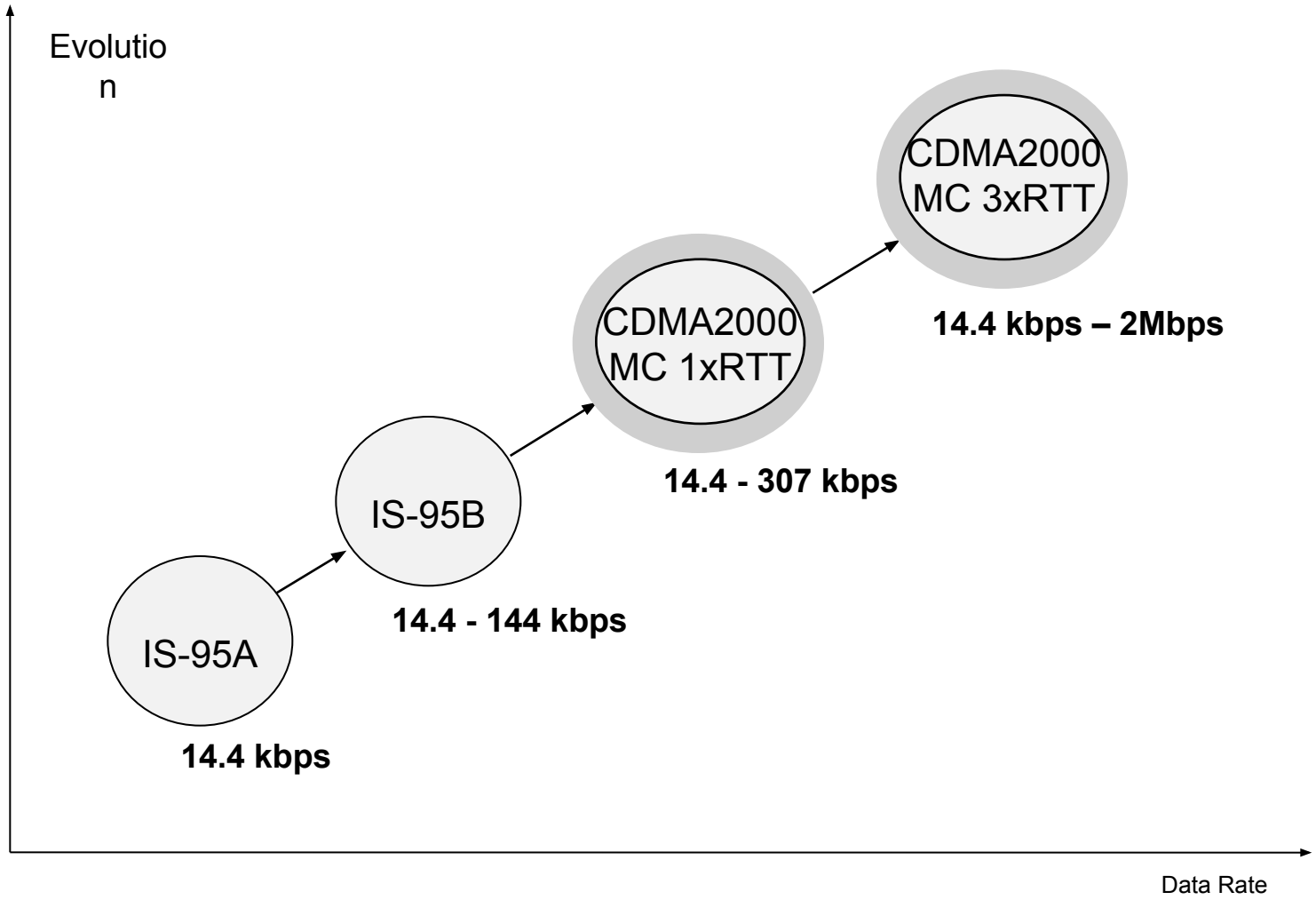
Digital Cellular Systems

Standard	Frequency Band	Multiple Access	Modulation	Channel BW
USDC	824-894 MHz	TDMA	π/4-DQPSK	30KHz
IS-95	824-894 MHz 1.8-2.0 GHz	CDMA	QPSK/ BPSK	1.25MHz
GSM	890-960 MHz	TDMA	GMSK	200KHz
PDC	810-1501 MHz	TDMA	Π/4-DQPSK	25KHz

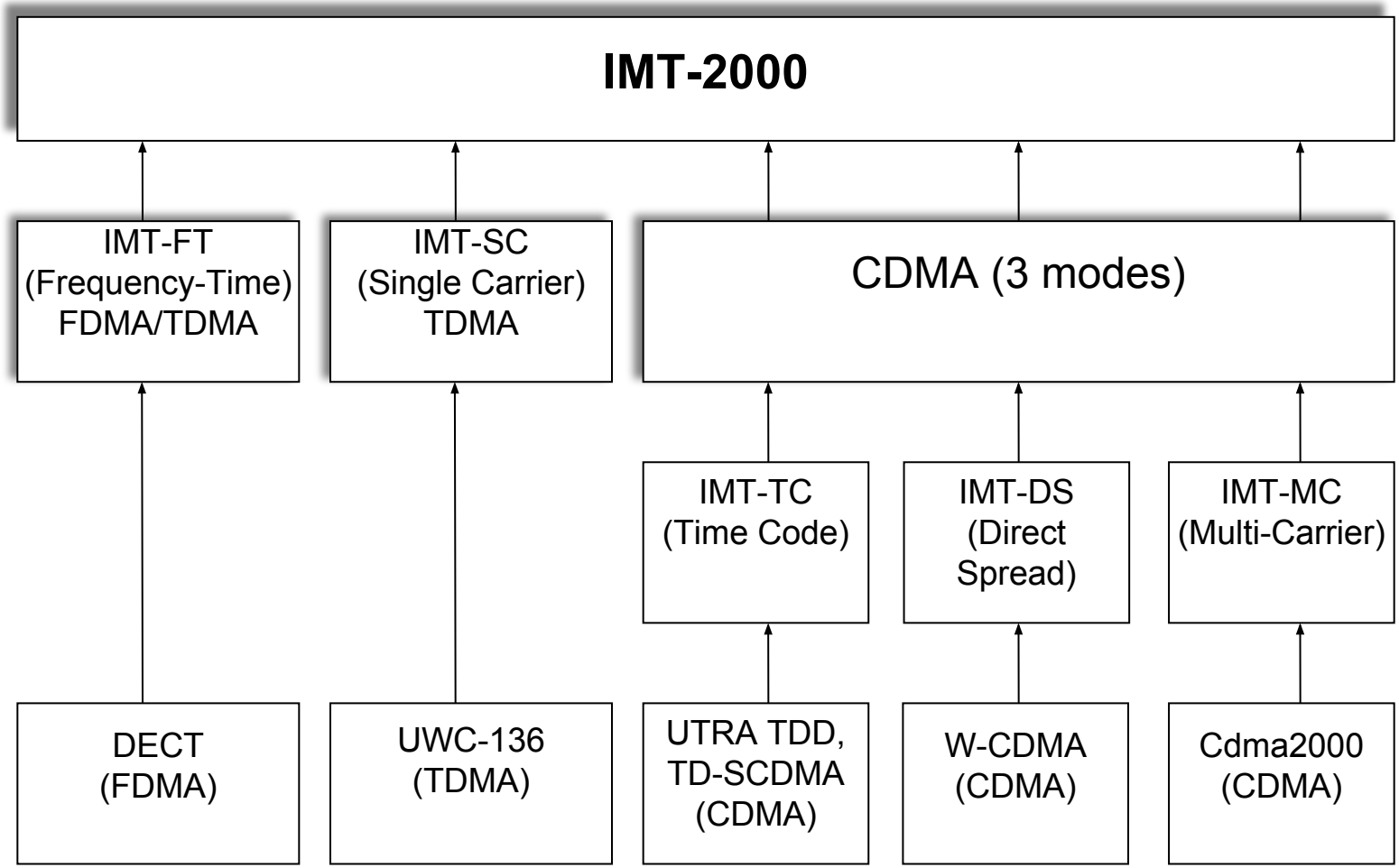
An Evolution Path from GSM to 3G Network



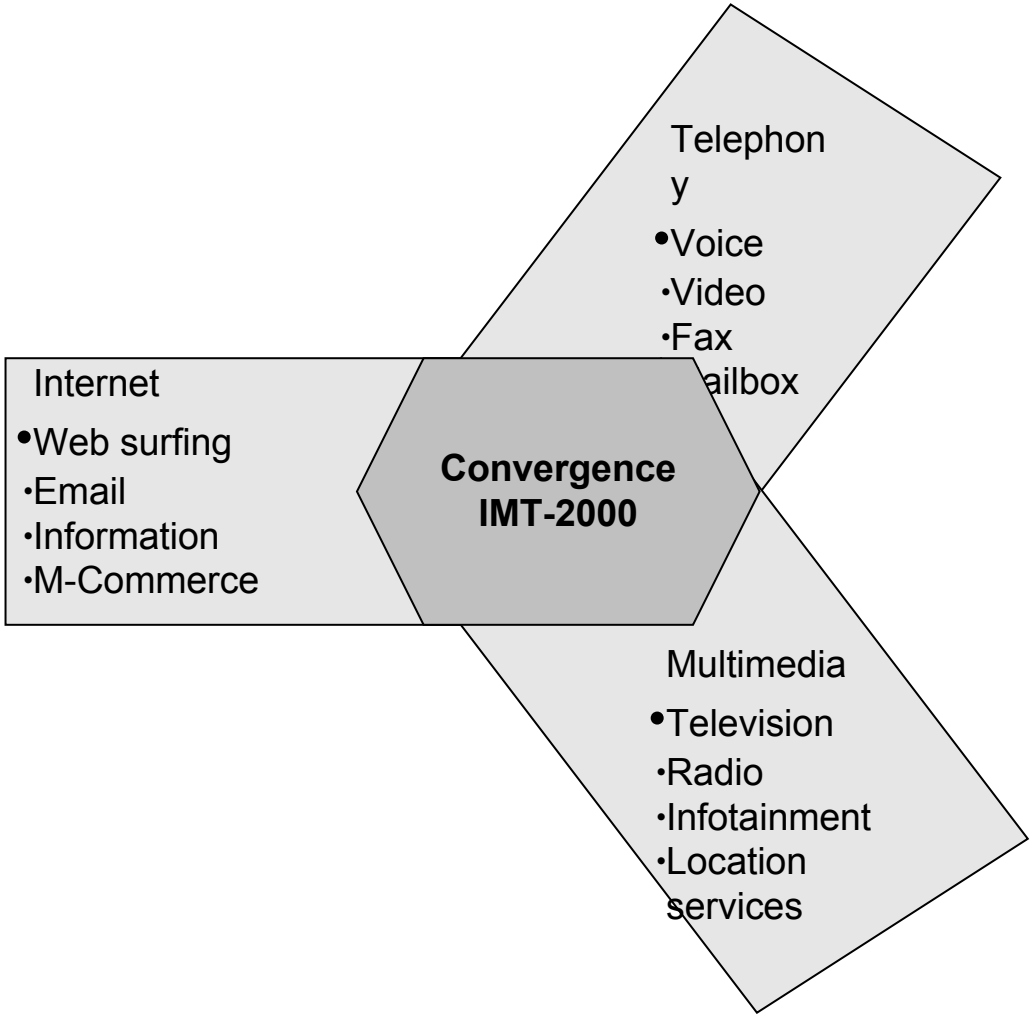
An Evolution Path from CDMA to 3G Network



Evolution of IMT-2000 standards



Convergence of Services in IMT-2000



IMT 2000 Services

- Voice
- **Switched Data**
- Messaging
- **Multimedia Messaging Service (MMS)**
- Immediate Messaging
- **Medium, High, and Interactive Multimedia**
- Sending multimedia postcards

Second Generation Digital Cellular Systems

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Wireless Networking Technologies

- **Wireless Local Area Network (WLAN)**
- **Wireless Personal Area Network (WPAN)**
- **Wireless Meteropolitan Area Network (WMAN)**

Wireless Communication Systems

Three most commonly used household wireless communication systems are:

- ✔ **Paging System**
- ✔ Cordless Phone System
- ✔ **Cellular Telephone System**

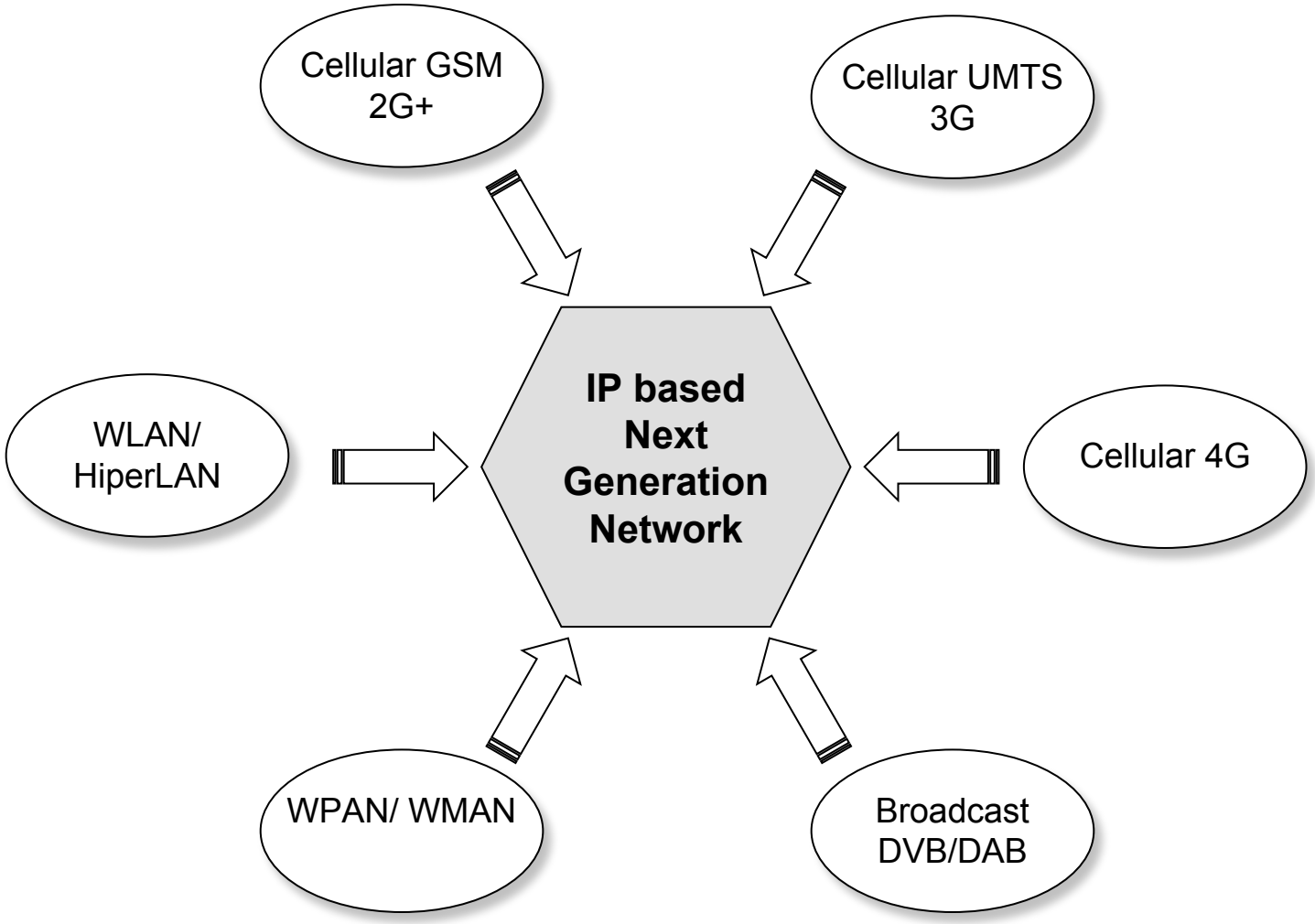
Comparison of Wireless Communication Systems

Service Type	Functionality	Operating Frequency	Level of infrast-ructure	Complexity	Hardware Cost	Range
Paging System	BS: Tx only MS: Rx only	< 1 GHz	High	BS: High MS: Low	BS: High MS: Low	High
Cordless Phone System	Transceivers	1 – 3 GHz	Low	BS: Low MS: Medium	BS: Medium MS: Low	Low
Cellular Phone System	Transceivers	< 2 GHz	High	High	BS: High MS: Medium	High

Cellular Communication Standards

Analog or Digital	Cellular Systems Standard	Frequency Band	Multiple Access Technique	Modulation Scheme	Region
Analog	AMPS	824-894 MHz	FDMA	FM	North America
Analog	NMT-900	890-960 MHz	FDMA	FM	Europe
Analog	ETACS	900 MHz	FDMA	FM	Europe
Analog	JTACS	860-925 MHz	FDMA	FM	Japan
Digital	USDC/IS-54 or 136	824-894 MHz	TDMA	π/4-DQPSK	North America
Digital	IS-95	824-894 MHz; 1.8-2.0 GHz	CDMA	QPSK/ OQPSK	North America
Digital	GSM	890-960 MHz	TDMA	GMSK	Europe
Digital	PDC	810-1501 MHz	TDMA	π/4-DQPSK	Japan

Vision of Next Generation Network



Next Generation Wireless Network

Functional Requirements:

- ✔ **Very high-speed and high-quality transmission**
- ✔ Open platform
- ✔ **Flexible and varied service functions**

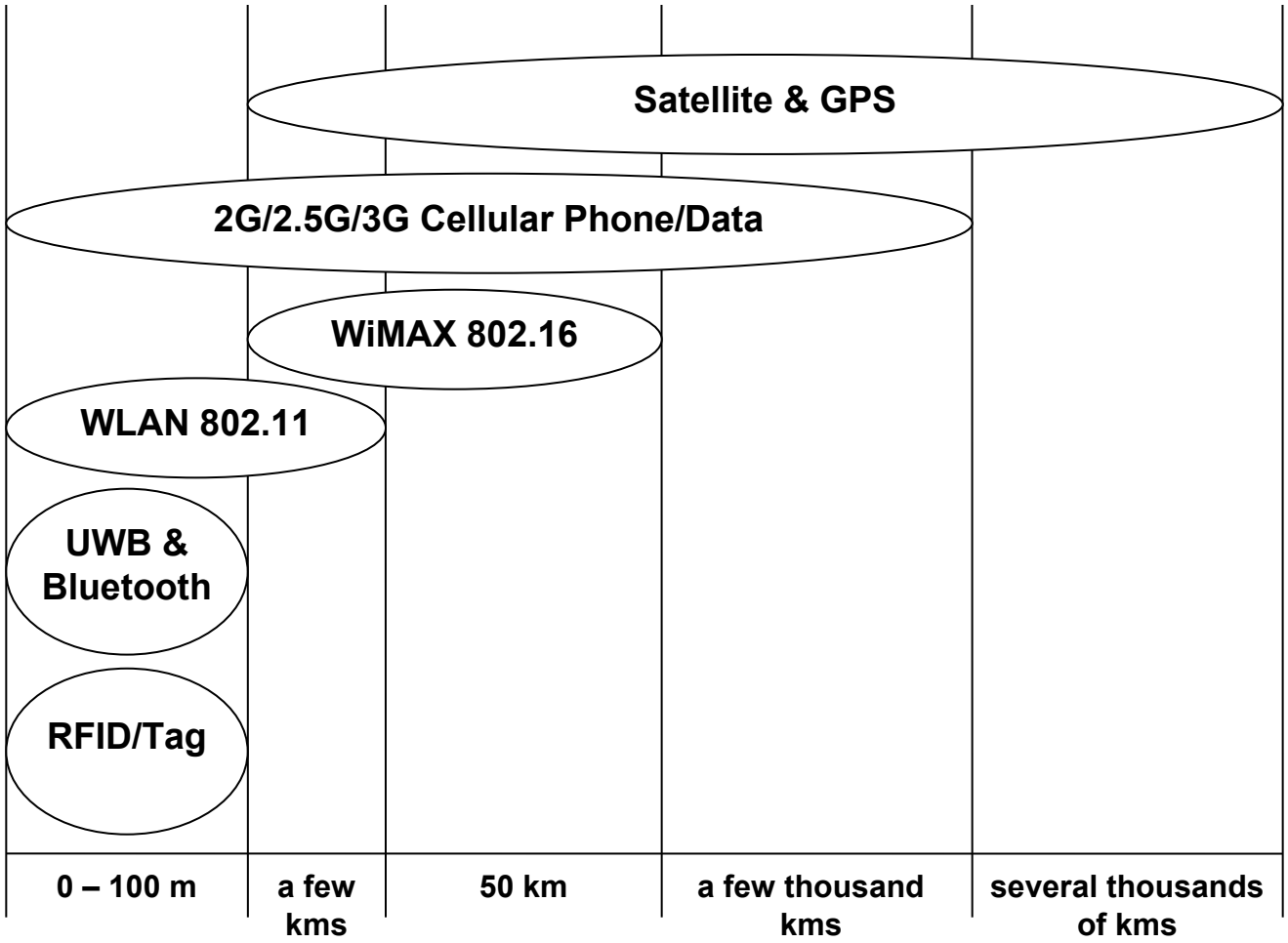
Comparison of Cellular Network Generations

First Generation	Second Generation	Third Generation	Next Generation
• **Analog Transmission**	• **Digital Transmission**	• **Digital Transmission**	• **Digital Transmission**
• Mainly Speech Communication	• Mainly Speech Communication	• Mainly Speech Communication	•Mainly Speech and Video Communication
• **Voice Band Data**	• **Digital Data**	• **Increasing Digital Data**	•**Mainly Digital Data**
• Circuit Switched	• Circuit Switched	• Increasingly Packet Switched	• Mainly Packet Switched
• **Local Systems**	•**Global Roaming**	•**Global Roaming**	•**Global Roaming**

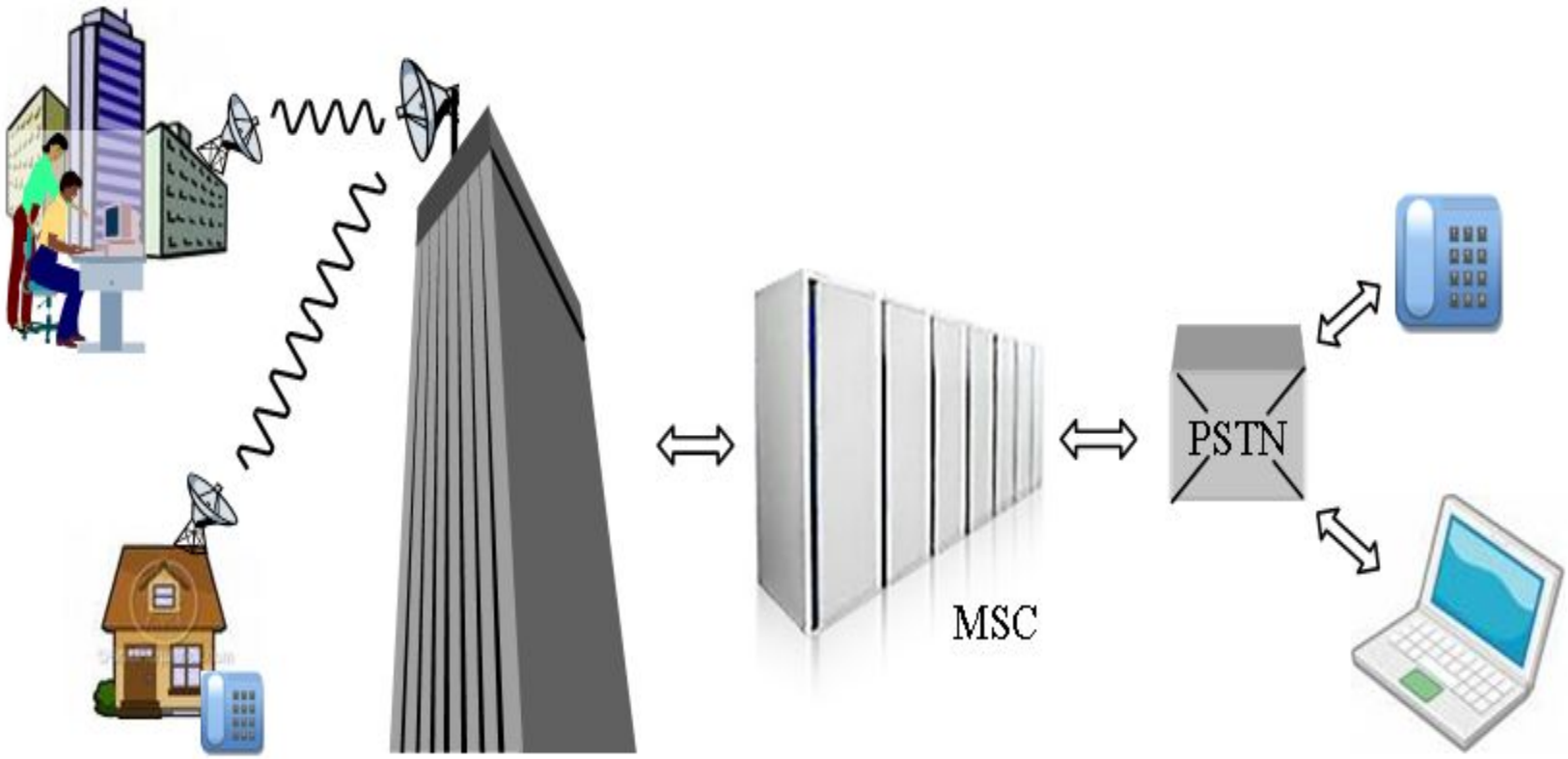
Wireless Data Communications Technologies

S. No.	Wireless Technology	Data Speed	Radio Coverage
1.	2G digital cellular	10 kbps	Nationwide through roaming
2.	2.5G digital cellular	Up to 384 kbps	Nationwide through roaming
3.	3G digital cellular	Up to 2 Mbps	Nationwide through roaming
4.	WLAN 802.11b	11 Mbps	Building/campus, 100-120 meters
5.	WLAN 802.11g	54 Mbps	Building/campus, 90 meters
6.	WMAN 802.16 WiMax	75 Mbps	Metro city area, 56 kilometers
7.	Bluetooth	1 Mbps	Room/House, within 10 meters
8.	Ultra Wide Band (UWB)	100 Mbps	Auditorium, 50 meters
9.	RFID	Few kbps	Small area within a store room, 2.5 cm to 100 m
10.	GPS and Satellites	250 ms delay	Worldwide global

Wireless Data Communication Technologies



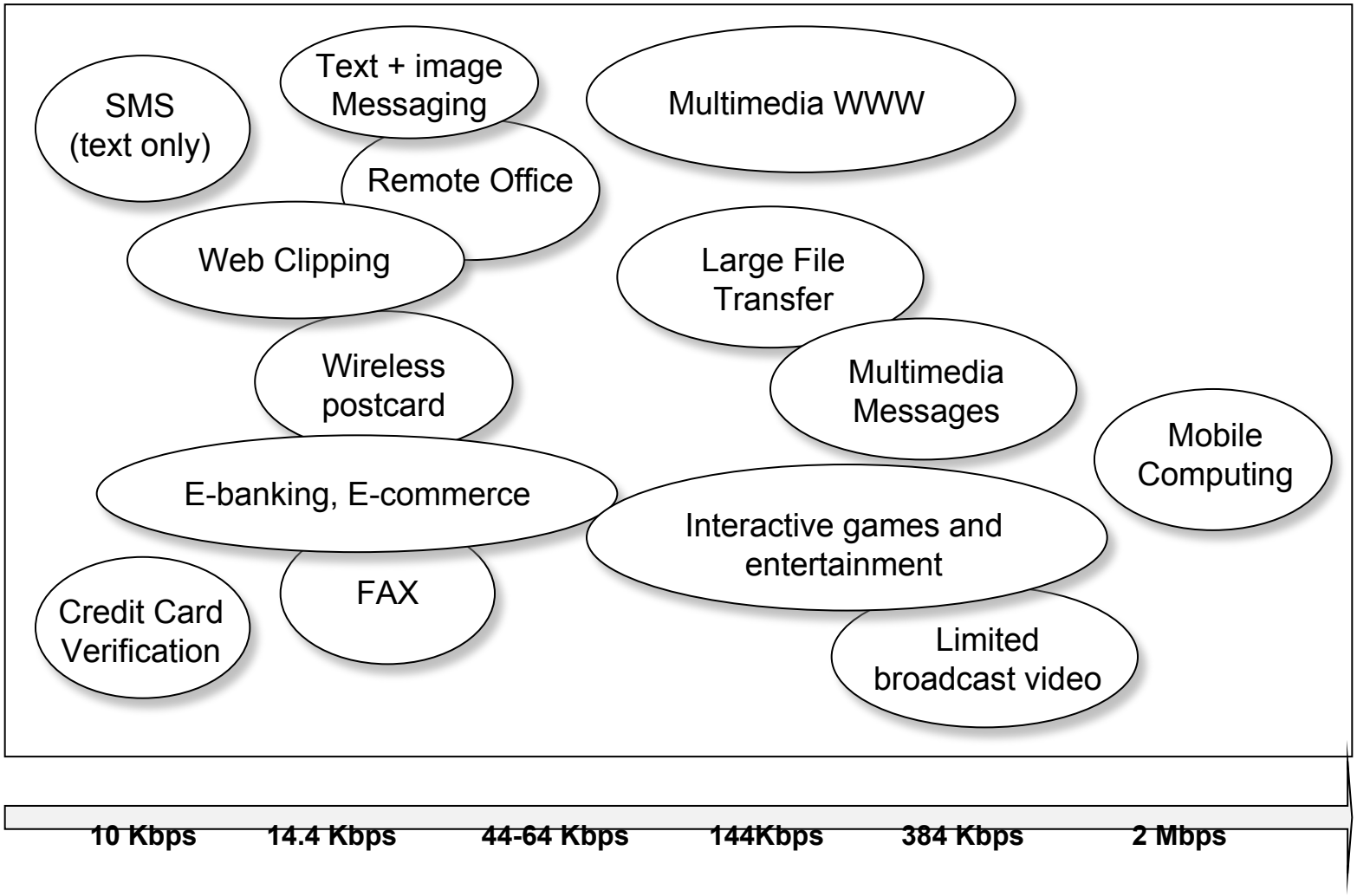
A Typical Fixed Wireless Network



Applications of Wireless Communications

- Office and household environments
- **Industrial control**
- Education sector
- **Health services**
- Government and military operations
- **Event and travel management**
- Home entertainment
- **Environmental and industrial research**

Potential Market Areas and Data Rates



Target Business Areas

- The Automotive Industry Market
- **The Fleet Management**
- Vehicle Positioning Market
- **The Utilities Market**
- The Security Systems Market
- **Vending Machines**

Challenges for Research

Wireless communications – a major technological areas for research as well as industrial applications.

True wireless multimedia services – required by highly mobile subscribers seamlessly on a global arena.

Diverse IP multimedia applications.

Summary

- Cellular Mobile Communications
- **Remote Wireless Internet Connections**
- Wireless Networks
- **Mobility, increased network reliability, easier and less expensive installation, and support for disaster recovery**

Thanks!